The Effects of School Principals on Teachers, Classrooms, and Families

October 2026

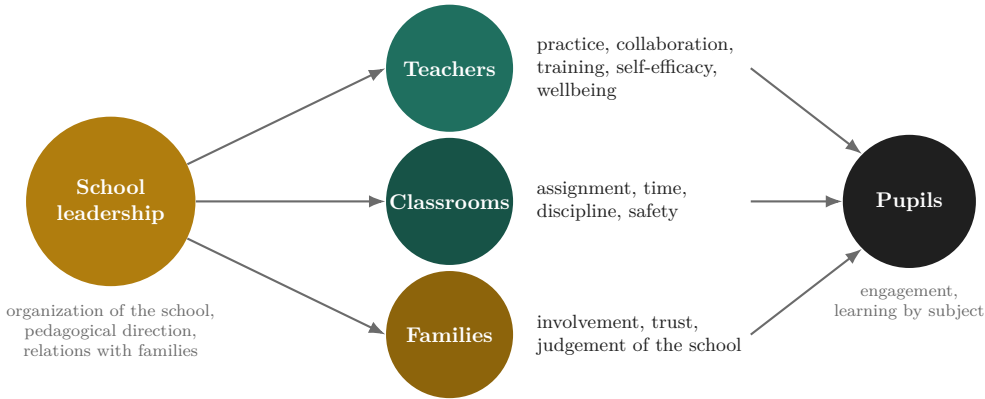
Through which inputs do school principals affect learning, and how much do they matter?

Related literature

- One SD of principal quality moves achievement by 0.03 to 0.10 SD (Branch, Hanushek, and Rivkin, 2012; Bartanen, Husain, and Liebowitz, 2024).
- A change of principal lowers achievement by 0.03 SD and raises teacher exit (Bartanen, Grissom, and Rogers, 2019).
- The identified channel is personnel: who is hired, retained, and dismissed (Grissom and Bartanen, 2019; Muñoz and Prem, 2024).

What is not identified is the effect on the practice and effort of a fixed staff, on the organization of classrooms, and on parental inputs.

Causal map



Exogenous variation enters at the leadership node. Every other node is observed in the data.

Institutional setting

- Spanish principals are teachers elected for four years. They do not hire, dismiss, set pay, or set the curriculum.
- A change of principal therefore changes leadership on a fixed staff.
- Terms end by rule: retirement at 65, and at most two renewals, twelve years.

Data: the universe of Madrid public primary schools, 2015–2018, with principals, teachers, timetables, every pupil and parent by classroom, and test scores by subject.

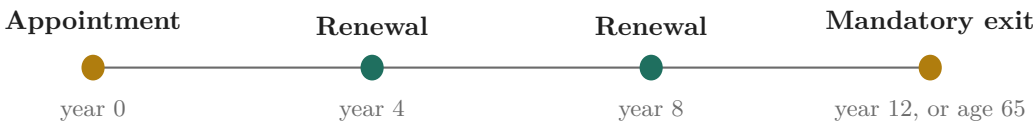
The selection of principals, Madrid 2015–2018

	Share
Competitions with a single candidate	0.75
Competitions with two or more candidates	0.25
Deserted competitions, filled by the administration	0.04
Contested competitions not won by the top-merit candidate	0.39

Source: official records of about 1,000 competitions, with candidates, merit scores, winners, and committees (Montalbán, 2026).

The incoming principal is not selected on quality. The parameter is the effect of the manager the system produces, and the records describe who that is.

Identification strategy



- Treatment: an exit forced by age or tenure. Control: schools whose principal has not reached either limit.
- Assumption: the exit date is unrelated to school shocks. Test: no pre-trends.
- Voluntary and deserted successions estimated separately, as the selection benchmark.

Empirical specification

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

- τ_s: first school year under the new principal. i: a teacher, classroom, pupil, or parent.
- The school is the unit of treatment; outcomes are repeated cross-sections within it.
- Staggered-robust estimator, not-yet-treated controls, clustered by school. 2SLS on the realized change where the tenure rule does not bind.

Dispersion and heterogeneity of principal effects

Dispersion of principal effects, from excess variance around rule-based exits (Jones and Olken, 2005):

$$\text{Var}(\Delta Y_{st} \mid \text{exit}) - \text{Var}(\Delta Y_{st} \mid \text{no exit}) = 2\sigma_\theta^2$$

Heterogeneity by characteristics determined before the change:

$$Y_{ist} = \alpha_s + \lambda_t + \beta D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

W_s includes how the vacancy was filled, the successor's merits and age relative to the incumbent, the successor's practices as a teacher in the previous year, and the staff's rating of the incumbent.

Challenges

- Power: 100 to 200 mandatory exits in primary over three transitions; effects of 0.03–0.05 SD on scores, larger on survey indices.
- Panels: the share of tutors and pupils re-observed is unknown.
- The tenure rule needs appointment histories since 2004; the age rule needs the count of principals above 65.

Next steps

- Count exits by age and by tenure among primary schools, by year.
- Share of tutors re-observed across years.
- Re-run the deserted-competition regression with mandatory exits as treatment and the incumbent's prior satisfaction as outcome: a flat pre-period is the pre-trend test.

Appendix: recent work on school and public-sector managers

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

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Branch, G. F., E. A. Hanushek, and S. G. Rivkin. 2012. "Estimating the Effect of Leaders on Public Sector Productivity: The Case of School Principals." NBER Working Paper 17803.

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Muñoz, P., and M. Prem. 2024. "Managers' Productivity and Recruitment in the Public Sector: The Case of School Principals." *American Economic Journal: Economic Policy* 16(4): 223–253.